Polarization by Aligned Grains: What SEDust and MoCafe Would Need

Abstract

This memo audits what it would take to replace the empirical dichroic polarization treatment of Seon (2018, ApJ 862, 87) with genuine aligned-grain physics in the current SEDust + MoCafe stack. The 2018 model is dissected first: its *scattering* polarization was already exact (full numerical Mie Mueller matrix with Stokes tracing), but its *dichroic* polarization was an add-on fitting formula with no alignment physics, no wavelength dependence, and additive rather than multiplicative accumulation of Stokes Q and U. The three ingredients that would be needed instead are then examined in turn: polarized cross sections, an alignment prescription, and a polarized transfer solver. The central practical finding is that the first of these is already in hand. The file `q_DH21Ad_P0.20_Fe0.00_1.400.dat.gz` stores three separate orientation blocks rather than an orientation average, so $C_{\rm pol}$ can be built today without any new T-matrix computation; values derived from it reproduce the Hensley & Draine (2023) released `polarized_extinction.dat` to four significant figures. A staged plan follows, in which stage 1 (SEDust cross sections and an emission accumulator) is a few days of low-risk work, stage 2 already permits a quantitative comparison against the SOFIA/HAWC+ 154 μm maps of NGC 891, and only stage 4 (self-consistent alignment) requires the invasive change of a vectorial radiation-field pass. Validation targets and the points that remain unverified are listed explicitly.

Contents

1. Motivation: anatomy of the 2018 model

The question this memo answers is narrow: if Seon (2018, ApJ 862, 87; `2018ApJ...862...87S`), "Polarization as a Probe of Thick Dust Disks in Edge-on Galaxies: Application to NGC 891", were redone now, what would have to be built? Answering it requires an exact statement of what that model did and did not contain. The description below is taken from arXiv:1806.06101v1.

1.1 What was already correct: scattering polarization

The scattering half of the 2018 calculation was not an approximation. Spherical grains were treated with the full numerical Mie Mueller matrix, retaining all four independent elements S_{11}, S_{12}, S_{33}, S_{34}; photons carried a Stokes vector; the scattering and peel-off bookkeeping followed the algorithm of Peest et al. (2017, A&A 601, A92; `2017A&A...601A..92P`); and the implementation was verified against the polarized scattering benchmark of Bianchi, Ferrara & Giovanardi (1996, ApJ 465, 127; `1996ApJ...465..127B`). No Henyey–Greenstein or Rayleigh proxy was used for the phase matrix. That part of the code needs no revision.

1.2 What was empirical: dichroic polarization

The dichroic half descends from the prescriptions of Wood (1997, ApJ 477, L25; `1997ApJ...477L..25W`) and Wood & Jones (1997, AJ 114, 1405; `1997AJ....114.1405W`). Seon (2018) adopted the Whittet et al. (2008, ApJ 674, 304; `2008ApJ...674..304W`) empirical relation $P_K/\tau_K \approx 5.08\,\tau_V^{-0.52}$ and converted it to

$$P_V = 2.7\,\tau_V^{0.48} \quad [\%], \tag{1}$$

then, at every interaction site, added a polarized fraction $P_V \sin^2\theta$ oriented parallel to the local magnetic field projected on the sky, with θ the angle between the field and the photon direction. The author's own summary in that paper is unambiguous: "we assumed spherical dust grains when calculating the Mueller matrix and adopted an empirical approach to take into account the dichroic polarization effect."

The internal tension is worth stating plainly. Under a strictly spherical grain assumption the dichroic cross-section difference vanishes identically, so the dichroic term was necessarily grafted on from outside the optical model rather than derived from it.

1.3 What was absent: alignment physics

Radiative torque (RAT) alignment appears in the 2018 paper only as introductory background. Neither the Rayleigh reduction factor R nor an alignment size threshold $a_{\rm align}$ enters the calculation anywhere. All alignment efficiency is absorbed into the single exponent 0.48 and the normalization of Eq. (1), which are Milky Way averages. That is a double-counting hazard as well as a physics gap: the Whittet relation already folds in turbulent depolarization along real Galactic sight lines, plus a mean alignment efficiency and a mean grain composition, and the model then imposes its own explicit field geometry on top of that same quantity.

The magnetic field was treated as two separate deterministic runs, purely toroidal and purely vertical, compared against each other. There was no turbulent component and no field-strength parameter. Crucially, the field in that model sets only the polarization position angle and the $\sin^2\theta$ projection factor; it has no influence on alignment efficiency, because no alignment efficiency is computed.

The dust model was MRN with Draine & Lee (1984, ApJ 285, 89; `1984ApJ...285...89D`) and Draine (2003, ApJ 598, 1017; `2003ApJ...598.1017D`) silicate and graphite optical constants, with PAHs excluded. The choice matters more than it looks: the maximum single-scattering polarization is 36% for that mixture but 24% for a WD01 mixture, a factor of 1.5. Only the V band was computed; far-infrared and submillimeter polarized *emission* was not a topic of the paper at all.

1.4 The implementation that survives in the tree

The transport code was MoCafe with forced first scattering and peeling-off. Dichroism was not applied by solving an extinction matrix; polarization was simply incremented at each interaction site. Two consequences follow directly: dichroic extinction never feeds back into the optical depth, and birefringence (the $Q,U \leftrightarrow V$ conversion) is absent entirely.

The actual code is still present at `~/MoCafe/Galaxy_pol/dichroic.f90`. Lines 75–89 read:

```
select case(par%dichroic_method)
   case (2)
      pol = par%pol_dichroic * tau**(3.0_wp/4.0_wp)
   case (3)
      pol = par%pol_dichroic * tau**(0.48_wp)
   case (1)
      pol = par%pol_dichroic * tau**(0.48_wp) * sint**2
   case default
      pol = par%pol_dichroic * tau**(3.0_wp/4.0_wp) * sint**2
end select
T1 = cos2p * pol
T2 = sin2p * pol
photon%Q = photon%Q + T1
photon%U = photon%U + T2
```

with the default amplitude `pol_dichroic = 0.013` set in `Galaxy_pol/define.f90:210`. Two distinct physical defects are visible here.

(i) The τ dependence is a fit, and the wavelength dependence is missing. The correct form for a uniform slab of aligned grains is set by the polarized optical depth $\Delta\tau = n_d C_{\rm pol} s$, giving

$$p = \tanh(\Delta\tau), \tag{2}$$

which reduces to $p \simeq \Delta\tau$ only in the optically thin limit. Neither $\tau^{0.75}$ nor $\tau^{0.48}$ is that function. More seriously, $C_{\rm pol}$ carries the entire wavelength dependence of dichroism, and it is absent from the expression above: this formulation cannot reproduce a Serkowski curve even in principle, because it has no λ argument.

(ii) The accumulation is additive where the physics is multiplicative. The extinction matrix mixes I into Q, so the correct operation is a matrix exponential acting on the Stokes vector. Incrementing Q and U by a fixed amount at each site agrees with that only while $\tau \ll 1$; the two diverge once $\tau \gtrsim 1$, which is precisely the regime a thick edge-on disk model is built to probe.

2. What has changed since 2018

Observations. NGC 891 now has far-infrared polarimetry. SOFIA/HAWC+ observed it at $154\,\mu$m (Jones et al. 2020, AJ 160, 167; `2020AJ....160..167J`), and a deeper analysis was published by Kim et al. (2023, AJ 165, 223; `2023AJ....165..223K`). The conclusion of the latter is directly in tension with the 2018 setup: the low polarization fraction observed in the disk, together with its narrow dispersion, requires many turbulent decorrelation cells along each line of sight. A smooth, purely toroidal or purely vertical field cannot produce that combination. Any redo therefore needs a turbulent field component as a first-class ingredient, not an afterthought.

Methodology. Two papers have since addressed exactly the technical obstacles the 2018 implementation stepped around. Baes et al. (2019, A&A 630, A61; `2019A&A...630A..61B`) analyzed the optical-depth problem in a dichroic medium, i.e. how to sample path lengths when the extinction coefficient depends on the polarization state being transported. Peest et al. (2023, A&A 673, A112; `2023A&A...673A.112P`) provided a full 3D Monte Carlo radiative transfer framework for aligned spheroidal grains, together with analytic benchmarks. Both are directly usable.

3. Polarized optical properties

3.1 The minimum required set

Three cross sections beyond the scalar ones are needed: $C_{\rm ext}^{\|}$ and $C_{\rm ext}^{\perp}$ for dichroic extinction, $C_{\rm abs}^{\|}$ and $C_{\rm abs}^{\perp}$ for polarized thermal emission, and $C_{\rm circ}$ for birefringence (the phase retardation that converts linear into circular polarization). In practice these collapse into the combinations $C_{\rm ext}$, $C_{\rm pol}$, $C_{\rm circ}$ defined below.

3.2 Modified picket fence approximation

The modified picket fence approximation (MPFA) of Draine & Hensley (2021b, ApJ 919, 65; `2021ApJ...919...65D`, Eqs. 11–17) reduces the orientation dependence to two reference orientations. For an oblate spheroid with symmetry axis $\hat{a}$, writing Θ for the angle between the electric field and $\hat{a}$,

$$C_E(\Theta) \;\simeq\; \cos^2\Theta\, C_E(0) + \sin^2\Theta\, C_E(90), \tag{3}$$

and identically for the magnetic-field-referenced C_H. Hensley & Draine (2023, ApJ 948, 55; hereafter HD23, `2023ApJ...948...55H`) then write the random-orientation and polarized cross sections as

$$C_{\rm ran} = \tfrac{1}{3}\left[C_E(0) + C_E(90) + C_H(90)\right], \tag{4}$$

$$C_{\rm pol}^{\rm oblate} = \tfrac{1}{2}\left[C_H(90) - C_E(90)\right], \tag{5}$$

their Eqs. 2 and 3 respectively. The accuracy of Eq. (3) is excellent for $2\pi a/\lambda \lesssim 1$ and degrades to roughly 10% near $a \sim \lambda$, but after integration over a realistic size distribution the residual error is a few percent (DH21b).

3.3 Partial alignment and the Rayleigh reduction factor

Partial alignment enters through a single scalar, the Rayleigh reduction factor

$$R \equiv \tfrac{3}{2}\langle\cos^2\beta\rangle - \tfrac{1}{2}, \tag{6}$$

which is the same quantity HD23 call $f_{\rm align}$. With θ the angle between the magnetic field and the line of sight, the partially aligned cross sections are

$$Q_{\rm abs}(\theta) = \langle Q\rangle + R\,(Q_{\|} - Q_{\perp})\left(\tfrac{1}{3} - \tfrac{\sin^2\theta}{2}\right), \tag{7}$$

$$Q_{\rm abs,pol}(\theta) = -R\,(Q_{\|} - Q_{\perp})\,\tfrac{\sin^2\theta}{2}. \tag{8}$$

The structural point for implementation is that the entire θ dependence factorizes into the single function $\sin^2\theta$. Only two wavelength-by-size tables need to be stored, the orientation-averaged one and the polarized one; the geometry factor is applied at run time in the transport loop. No angular dimension is added to the tables.

The corresponding observables are given by HD23 Eq. 6 for polarized extinction and HD23 Eq. 11 for polarized emission, the latter including the integral $\int dT\,(dP/dT)$ over the stochastic temperature distribution.

3.4 Sign convention

For an oblate spheroid $\hat{a}$ is the *short* axis, and alignment places $\hat{a} \parallel B$. Transmitted starlight is therefore polarized parallel to the plane-of-sky field $B_{\rm sky}$, while thermal emission is polarized perpendicular to it. Getting this backwards produces a 90° error that is easy to miss in an all-sky map and impossible to miss in a comparison against HAWC+ vectors.

3.5 The asset already in the tree

The single most useful finding of this audit is that no new T-matrix computation is required to get started. The file `astrodust/data/dielectric/q_DH21Ad_P0.20_Fe0.00_1.400.dat.gz` is not an orientation average. It contains three separate orientation blocks for each of $Q_{\rm ext}$, $Q_{\rm abs}$, $Q_{\rm sca}$, on the native 169 sizes × 1129 wavelengths grid. Its header states verbatim:

```
3 orientations: jori=1: k || a;; jori=2: k perp a, E || a;
              jori=3: k perp a, E perp a
```

so Eqs. (4) and (5) can be evaluated directly from it.

This has been verified numerically, not merely asserted. Values built from this table reproduce the HD23 released `polarized_extinction.dat` at 0.55 μm to four significant figures (ratio 1.0000), and a fit of the alignment function to the same data returns $a_{\rm align} = 0.07492\,\mu$m, $\alpha = 1.799$, $f_{\rm max} = 1.0$, identical to HD23 Table 1.

3.6 Index convention traps

The orientation index ordering is not consistent across Draine's own distributions, and this is a live source of silent sign errors. Draine & Fraisse (2009, ApJ 696, 1; `2009ApJ...696....1D`) use one ordering; the astrodust table uses another. Table 1 gives the mapping.

Table 1: Orientation index conventions. The same three physical orientations are numbered differently in two files from the same source.

Source	$k \perp a,\, E \parallel a$	$k \perp a,\, E \perp a$	$k \parallel a$
DF09 (`2009ApJ...696....1D`)	$j = 1$	$j = 2$	$j = 3$
`q_DH21Ad_*`	`jori` = 2	`jori` = 3	`jori` = 1

A third convention exists in the same tree: in `qlib_gra_D16MGemt_1.400`, `jori = 1` is the *random-orientation average*, not $k \parallel a$. This is documented in the code at `astrodust/sed/src/q_graphite_d16.f90:23--26`. Any reader routine must therefore dispatch on the file identity, never on the index alone.

3.7 HD23 alignment assumptions

For reference, the alignment model adopted by HD23 is: oblate spheroids with $b/a = 1.4$, porosity 0.20, and

$$f_{\rm align}(a) \;=\; \frac{f_{\max}}{1+(a_{\rm align}/a)^{\alpha}}, \qquad a_{\rm align} = 0.0749\,\mu{\rm m}, \quad \alpha = 1.80, \quad f_{\max} = 1.0, \tag{9}$$

with zero polarized contribution from PAHs. The axial ratio $b/a = 1.4$ is not a free aesthetic choice: it is the minimum departure from sphericity that the observed starlight polarization efficiency integral of DH21b will tolerate.

4. What alignment theory hands to radiative transfer

4.1 The interface is one number

Alignment theory, however elaborate, delivers exactly one scalar per grain radius to the transfer solver: $R(a)$. It factorizes into internal alignment (the angle β between the short axis $\hat{a}_1$ and the angular momentum J) and external alignment (the angle ξ between J and B),

$$R \;=\; \left\langle \frac{3\cos^2\beta - 1}{2} \cdot \frac{3\cos^2\xi - 1}{2} \right\rangle, \tag{10}$$

following Hoang et al. (2024, ApJ 965, 183; `2024ApJ...965..183H`, Eq. 5). The practical reduced form used in that same paper (their Eq. 12) is a two-state model,

$$R = \begin{cases} 0, & a < a_{\rm align}, \\ f_{\text{high-J}}\, Q_X^{\text{high-J}} + (1 - f_{\text{high-J}})\, Q_X^{\text{low-J}}, & a \geq a_{\rm align}, \end{cases} \qquad Q_X^{\text{high-J}} = 1, \quad Q_X^{\text{low-J}} \simeq 0.3. \tag{11}$$

4.2 The analytic alignment radius

The reference analytic chain for $a_{\rm align}$ is Hoang, Tram & Lee (2021, ApJ 908, 218; `2021ApJ...908..218H`). The RAT efficiency is approximated by

$$\bar{Q}_\Gamma \simeq \begin{cases} 8\,(\bar{\lambda}/a)^{-3}, & a \lesssim \bar{\lambda}/2.7, \\ 0.4, & a \gtrsim \bar{\lambda}/2.7, \end{cases} \tag{12}$$

the infrared rotational damping coefficient by

$$F_{\rm IR} \simeq 0.038\, U^{2/3}\, a_{-5}^{-1}\, n_3^{-1}\, T_1^{-1/2}, \tag{13}$$

and the suprathermal rotation condition $\omega_{\rm RAT} = 3\,\omega_T$ then yields

$$a_{\rm align} \simeq 0.055\,\hat{\rho}^{-1/7} \left(\frac{n_3 T_1}{\gamma_{-1} U}\right)^{2/7} \left(\frac{\bar{\lambda}}{1.2\,\mu{\rm m}}\right)^{4/7} (1+F_{\rm IR})^{2/7}\ \mu{\rm m}. \tag{14}$$

Inside a cloud the local field is reddened and attenuated, which is captured by

$$U = \frac{U_0}{1+0.42A_V^{1.22}}, \qquad \bar{\lambda} = \bar{\lambda}_0\left(1+0.27A_V^{0.76}\right), \qquad \bar{\lambda}_0 = 1.3\,\mu{\rm m}. \tag{15}$$

4.3 Rotational disruption

Rotational disruption (RATD) removes large grains whose spin stress exceeds the tensile strength $S_{\rm max}$. The important structural point is that disruption defines a *band*, not a cutoff: there is an upper bound $a_{\rm disr,max}$ as well as $a_{\rm disr}$, because sufficiently large grains again spin too slowly to be disrupted. Implementing RATD as a simple truncation of the size distribution is wrong.

4.4 Magnetically enhanced alignment and its 2025 revision

Magnetically enhanced RAT alignment (MRAT) is parameterized by the magnetic relaxation ratio $\delta_{\rm mag}$. Giang et al. (2023) adopt a step function, $f_{\text{high-J}} = 0.25$, 0.5, 1 for $\delta_m < 1$, 1–10, and > 10 respectively. That last branch is the one to be careful with.

Hoang (2025, ApJ 994, 115) revises the criterion to

$$\delta_{\rm mag,cri} \;=\; (1+F_{\rm IR})\,\frac{2-q^{\rm max}}{q^{\rm max}}, \tag{16}$$

showing that the familiar "$\delta_m > 10$ implies perfect alignment" rule was the special case $F_{\rm IR} \ll 1$. For $U \sim 10^3$ one has $F_{\rm IR} \sim 60$, so the required $\delta_{\rm mag}$ rises by roughly the same factor. Any implementation that keeps the 2016-era step function will therefore overestimate alignment in strong radiation fields, exactly where the polarized emission is brightest. The same paper finds that the attainable maximum of $f_{\text{high-J}}$ is about 0.45 for superparamagnetic grains and about 0.22 for ordinary paramagnetic grains, so the value 1.0 used by POLARIS is optimistic. It also proposes the environment classifier $U/(n_1 T_2)$: values ≤ 1 are collision-dominated, values > 1 are radiation-dominated, in which case the slow-rotation branch has $f_{\text{high-J}}^{\rm slow} = 0$.

Note the division of labor between these references: the authoritative closed-form expressions for $a_{\rm align}$ and $a_{\rm disr}$ remain those of Hoang et al. (2021). Hoang (2025) does not supply new closed forms; it recasts the problem in a dimensionless parameter space.

4.5 Angle convention

In this literature γ denotes the angle between the *mean magnetic field and the line of sight*, not a plane-of-sky angle. Polarization is maximal at $\gamma = 90°$. Mixing this up with a position angle is a common and quiet error.

4.6 Which axis do grains align with?

Everything above assumes the alignment axis is B. That holds only while Larmor precession is the fastest precession the grain undergoes. Lazarian & Hoang (2019, `2019ApJ...883..122L`) set the comparison out explicitly. A grain with a Barnett magnetic moment precesses about B on a timescale (their Eq. 4)

$$t_{\mathrm{L}} \; = \; \frac{4\pi}{5}\frac{g_e\mu_B}{\hbar}\,\rho_s\, s^{-2/3}\, a_{\mathrm{eff}}^2 B^{-1}\chi(0)^{-1} \;\approx\; 1.3\,\hat{\rho}\,\hat{s}^{-2/3}\, a_{-5}^2\,\hat{B}^{-1}\hat{\chi}^{-1}\;\;\mathrm{yr}, \tag{17}$$

with $\hat{B} = B/5\,\mu\mathrm{G}$ and $\hat{\chi} = \chi(0)/10^{-4}$. Anisotropic radiation drives a competing precession about k (their Eqs. 12–14),

$$t_{\mathrm{rad,p}} = \frac{2\pi}{\Omega_p}, \qquad \Omega_p = \frac{\bar{u}_{\mathrm{rad}}\,\bar{\lambda}\,a_{\mathrm{eff}}^2}{I_{\|}\,\omega}\,\gamma\,|\bar{Q}_\Gamma|, \tag{18}$$

$$t_{\mathrm{rad,p}} \approx 1.1\times 10^2\,\hat{\rho}^{1/2}\hat{s}^{-1/3}a_{-5}^{1/2}\hat{T}_d^{1/2}\left(\frac{u_{\mathrm{rad}}}{u_{\mathrm{ISRF}}}\right)^{-1}\left(\frac{\bar{\lambda}}{1.2\,\mu\mathrm{m}}\right)^{-1}\left(\frac{\gamma|\bar{Q}_\Gamma|}{0.01}\right)^{-1}\;\;\mathrm{yr}. \tag{19}$$

The criterion is simply which is shorter: if $t_{\mathrm{rad,p}} < t_{\mathrm{L}}$, "the radiation direction rather than that of magnetic field becomes the axis of alignment" and the grain is in the k-RAT regime rather than B-RAT.

Two consequences matter for the present purpose. First, the paper's own estimate is that both the radiative and the mechanical precession rates "are significantly smaller than the rate of Larmor precession," which "ensures that over most of the volume of the interstellar medium the alignment is happening in respect to magnetic field." For a galaxy-scale model such as NGC 891 the B-RAT assumption is therefore safe, and Eq. (17) versus Eq. (19) can be evaluated as a diagnostic rather than carried as a branch in the solver. The regime to watch is the immediate vicinity of bright sources, where $u_{\mathrm{rad}}/u_{\mathrm{ISRF}}$ is large enough to push $t_{\mathrm{rad,p}}$ below t_{L}.

Second, $t_{\mathrm{L}} \propto \chi(0)^{-1}$, so iron inclusions that raise the susceptibility also push the transition further away: the same magnetic enhancement that produces efficient MRAT alignment (§4.4) simultaneously protects the B-RAT assumption. Conversely, detecting a k-RAT signature would constrain $\chi(0)$, which is the observational program that paper proposes. Note also that Eq. (19) uses $|\bar{Q}_\Gamma| \approx 2.3(\lambda/a_{\mathrm{eff}})^{-3}$ for $\lambda > 1.8a_{\mathrm{eff}}$ and 0.4 otherwise, whose coefficient and transition point differ from the parameterization quoted in Eq. (12). The two are alternative fits to the same DDA results and agree to within tens of percent; use one consistently rather than mixing them.

4.7 The polarization hole

The mechanism is a chain of three steps, each of which the formulas above already contain: increasing A_V lowers U (Eq. 15), lower U raises a_{align} (Eq. 14), and once a_{align} exceeds the largest grain present, alignment vanishes and polarized emission goes to zero. The complete-loss condition scales as $A_{V,\max} \simeq 24.7(\cdots)$, evaluating to roughly 15.9, 6.9, and 3.0 mag for $n_{\mathrm{H}} = 10^4$, 10^5, 10^6 cm^{-3} with $a_{\max} = 0.25\,\mu$m. Grain growth pushes the transition to larger A_V, which is why the observed depth of the hole is a joint constraint on alignment and on growth rather than on either alone.

5. Polarized transfer and its implementation

5.1 The Stokes transfer equation

The transfer of the Stokes vector through an aligned medium is governed by a 4×4 extinction matrix. In the form given by POLARIS I (Reissl, Wolf & Brauer 2016, A&A 593, A87; `2016A&A...593A..87R`, their Eq. 7),

$$\frac{d}{ds}\begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix} = -n_d \begin{pmatrix} C_{\rm ext} & C_{\rm pol}\cos 2\varphi & C_{\rm pol}\sin 2\varphi & 0 \\ C_{\rm pol}\cos 2\varphi & C_{\rm ext} & 0 & C_{\rm circ}\sin 2\varphi \\ C_{\rm pol}\sin 2\varphi & 0 & C_{\rm ext} & -C_{\rm circ}\cos 2\varphi \\ 0 & -C_{\rm circ}\sin 2\varphi & C_{\rm circ}\cos 2\varphi & C_{\rm ext} \end{pmatrix} \begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix} + j, \tag{20}$$

where φ is the position angle of the projected field and j is the polarized emissivity.

Typographical error in the published matrix. The printed $(4,4)$ element in that paper is $C_{\rm pol}$; the correct entry is $C_{\rm ext}$, as written in Eq. (20) above. The reasoning is structural: V must suffer the same total extinction as I, Q, U, and only $C_{\rm circ}$, appearing antisymmetrically, may rotate U into V. The POLARIS source code implements it correctly, so this is an error in the paper rather than in the code.

5.2 How POLARIS assembles the matrix

The assembly happens in `DustMixture.cpp::calcEmissivityExt`, whose non-zero off-diagonal assignments are:

```
K[0][1] = K[1][0] = Cpol * cos2phi;
K[0][2] = K[2][0] = Cpol * sin2phi;
K[1][3] = Ccirc * sin2phi;
K[2][3] = -Ccirc * cos2phi;
K[3][1] = -Ccirc * sin2phi;
K[3][2] = Ccirc * cos2phi;
```

The cross sections themselves absorb the alignment factor in `DustComponent.cpp::calcCrossSections`, as

$$C_{\rm pol} = \tfrac{1}{2} R\,(C_{\rm ext}^{\perp} - C_{\rm ext}^{\parallel})\,\sin^2\theta, \tag{21}$$

$$C_{\rm pabs} = \tfrac{1}{2} R\,(C_{\rm abs}^{\perp} - C_{\rm abs}^{\parallel})\,\sin^2\theta. \tag{22}$$

A single scalar R thus scales the dichroic and the birefringent amplitudes simultaneously, which is the concrete expression of the statement in §4. that the alignment interface is one number.

5.3 Prefer the closed-form solution to adaptive integration

If the density and the alignment direction are constant within a cell, $\exp(-Kns)$ has a closed form (Peest et al. 2023, `2023A&A...673A.112P`): I and Q evolve through cosh and sinh of $C_{\rm pol}\,ns$, U and V through cos and sin of $C_{\rm circ}\,ns$, with an overall factor $e^{-C_{\rm ext}ns}$. POLARIS instead integrates Eq. (20) with an adaptive Runge–Kutta–Fehlberg 4(5) scheme, applying the error test to each of the four Stokes components with $\epsilon_{\rm err} = 10^{-8}$. For a cell-based grid the closed form is strictly better: it is exact, it is cheaper, and it removes a tolerance parameter. Peest et al. (2023) also publish analytic benchmark problems, which their MCpol implementation reproduces to within 0.1%; those benchmarks should be adopted as regression tests.

5.4 The three-pass structure and what it buys

POLARIS runs three passes, and the division of labor is the single most useful architectural lesson available:

1. Monte Carlo thermal equilibrium, with alignment ignored.
2. Monte Carlo radiation field, storing a *vector* u_λ and the anisotropy measure $\gamma_\lambda = |\sum u_i| / \sum |u_i|$.
3. Observables: alignment switched on, Stokes transfer along rays.

During the Monte Carlo passes the alignment is deliberately randomized (`DustMixture.cpp:652--656`). The consequence is worth stating explicitly, because it removes what looks like the largest obstacle: *polarized observables do not require the Monte Carlo transport itself to be polarized*. What is given up is the feedback of dichroic extinction on the radiation field, which is an approximation but a controlled one.

5.5 The radiation field must be vectorial

RAT torque depends on the anisotropy of the local radiation field, so a scalar mean intensity is insufficient: alignment simply cannot be computed from J_λ alone. What is required per cell and wavelength is the vector sum as well as the scalar one,

$$\bar{u}_\lambda = \sum_i |\vec{u}_{\lambda,i}|, \qquad \gamma_\lambda = \frac{\left|\sum_i \vec{u}_{\lambda,i}\right|}{\sum_i |\vec{u}_{\lambda,i}|} \in [0,1], \tag{23}$$

with γ_λ the anisotropy degree that enters the torque. This has a hard corollary. The modified random walk (MRW) destroys directional information by construction and therefore cannot be used in the pass that feeds alignment. It remains available in the thermal pass.

5.6 Torque efficiency tables are not needed

A pleasant surprise: POLARIS's `calcAlignedRadii` has the tabulated route (`getQrat`) commented out and uses the power-law approximation of Eq. (12) instead. Equivalent results are therefore obtainable without computing or storing a T-matrix torque-efficiency table, which removes what would otherwise be a substantial Phase-A style computation.

5.7 Stochastic heating and polarization

POLARIS supports the combination, using the Camps et al. (2015) formulation, and the structure is simple: the temperature probability weighting $P(T)$ multiplies I and Q,U identically. Two consequences follow. Within a single grain size the polarization fraction is *independent* of $P(T)$; the temperature distribution affects the net polarization only through the relative weighting of different sizes.

Physically the two populations barely overlap in any case. Alignment sets in near $a_{\rm align} \simeq 0.075\,\mu$m, whereas stochastic heating dominates for $a \lesssim 0.01$–$0.05\,\mu$m. The observational counterpart is the non-detection of polarization in the

3.4 μm C–H feature, 0.06% $\pm$ 0.13% (Chiar et al. 2006, ApJ 651, 268; `2006ApJ...651..268C`), consistent with the HD23 assumption of zero polarized PAH emission.

Nevertheless the correct treatment is essentially free here, and that is an argument specific to this code base. SEDust already solves $P(T)$; carrying the polarized emission through the same integral costs one additional accumulator. It is worth noting that the Hoang group papers all use equilibrium temperatures only; Tram et al. (2025, GRADE-POL) states explicitly that the temperature distribution was discarded. Doing it properly would be a genuine, if modest, improvement over the reference implementations.

6. Current state of the codes

6.1 SEDust

SEDust is in better shape than expected. The polarized cross-section table `q_DH21Ad` is already present with all three orientation blocks (§3.), and its correctness has been verified against the HD23 release. The `tmatrix/` directory links only `tmd_one.f`, the random-orientation average driver; the fixed-orientation code `ampld.lp.f` is present but unused. As long as the astrodust model is the target, no T-matrix run is needed at all; that code becomes relevant only for exploring other axial ratios or compositions.

The structural entry point is `grain_pop_t` in `sed/src/dust_model_mod.f90`: adding a `Cpol(NLAM,NA)` component there propagates naturally through the existing population loop. On the emission side, one polarized accumulator alongside the existing one in `sed_astrodust.f90` completes the change.

6.2 MoCafe v2.00

MoCafe already carries the polarization state. `photon_type` has `I,Q,U,V` (`define.f90:117`), there is a `use_stokes` switch (line 213), and `scatter_dust_stokes` interpolates S_{11}, S_{12}, S_{33}, S_{34}. SEDust is embedded through `dustemis_mod`, and the code passes the Camps et al. (2015) SHG benchmark. The task is already on record: the roadmap contains `[TODO, L] Polarized thermal emission from aligned grains. Long-term.`

Four things are missing:

1. No magnetic field exists anywhere (a source-wide grep returns zero hits). This is the largest single gap.
2. Extinction is scalar: `tau = rhokap * length`.
3. `jtally_mod` accumulates a *scalar* J_λ only, and must be vectorized before any self-consistent alignment calculation is possible.
4. There is no polarized emission source term.

6.3 Reference sources in the tree

Two POLARIS checkouts are available for reference: `Grain/POLARIS/` (branch `master-basic`, commit `03280f5`) and `Grain/NewPOLARIS/` (branch `main`, commit `e508e89`; tag `v1 = a4fcb4d` corresponds to Zenodo `10.5281/zenodo.14199014`, the published version, with `main` ahead only in the README).

NewPOLARIS is the MRAT and RATD extension of Giang, Hoang & Kim (2023). It is *not* merged upstream and was forked from an older layout (`Dust.cpp`/`Dust.h`), so a direct diff against the baseline is not possible; only the physics can be ported. What it adds: the `ALIG_MRAT`, `ALIG_kRAT`, `ALIG_INELASTIC` flags; the keywords <number_cluster>, <volume_filling_cluster>, <iron_fraction>, <change_f_highJ>, <inelascity>, <disrupt> RATD; and grid fields holding critical radii for each cell (`GRIDadisr`, `GRIDamaxJB_Lar`, `GRIDadg_lower/upper`), which moves the expensive physics into a preprocessing step. The stock POLARIS has no MRAT, and its `ALIG_KRAT` is a dead flag: it is connected to neither the parser nor any physics path.

7. Staged proposal

Table 2: Proposed stages. Stage 2 is the point at which quantitative comparison against SOFIA/HAWC+ becomes possible.

Stage	Scope	Effort	Content
1	SEDust cross sections	days, low risk	Add `Cpol` and $f_{\rm align}$; add a polarized emission accumulator.
2	MoCafe ray tracing	moderate	Polarized emission in ray tracing and peel-off, using the Peest closed form.
3	MoCafe forward MC	moderate	Dichroic extinction; keep scalar τ for path sampling and correct by weight.
4	Self-consistent alignment	large	Vectorial radiation-field pass; MRW disabled; memory $\times 4$.

Stage 1. Add `Cpol` and $f_{\rm align}$ to SEDust, together with a single polarized emission accumulator alongside the existing one. Nothing in the solver core needs to change. Validation proceeds in four steps: first reproduce the released `polarized_extinction.dat`, then check $p_{\rm submm}$ against Planck, then $R_{S/V}$, and finally the Serkowski parameters $\lambda_{\rm max}$ and K.

Stage 2. Polarized emission in the MoCafe ray tracing and peel-off paths, integrated with the closed-form solution of §5.. This is the milestone worth aiming at first: it already allows a quantitative comparison with the HAWC+ 154 μm maps, which did not exist in 2018.

Stage 3. Dichroic extinction in the forward Monte Carlo. The recommended approach keeps the scalar τ for path-length sampling and applies a weight correction, which sidesteps the circularity analyzed by Baes et al. (2019). At this stage the empirical `dichroic.f90` formula is replaced by Eq. (2) with a wavelength-dependent $C_{\rm pol}$.

Stage 4. Alignment computed internally. This requires the vectorial radiation-field pass, forbids MRW in that pass, and raises the radiation-field memory by a factor of four. Until then the sensible choice is the SKIRT approach: treat $f_{\rm align}(a)$ as an input function, with the HD23 fit of Eq. (9) as the default.

8. What the update actually touches

The stages above are stated in physical terms. This section translates them into the specific data structures and routines that would change, based on the state of the two trees as of 2026 July. Line numbers are given where they pin down a unique insertion point; they will drift.

8.1 SEDust, stage 1

Read the polarized cross sections. The existing loader `load_q_table` (`sed/src/q_table.f90`) reads the flat table produced by `run_tmatrix`, whose columns are `lambda a_eff Qext Qabs Qsca albedo g flag`. The HD21 file has a different shape: three orientation blocks, each 1129×169, written as `((Q(jw,jr,jori),jw=0,1128),jr=0,168),jori=1,3` for each of $Q_{\rm ext}$, $Q_{\rm abs}$, $Q_{\rm sca}$ in turn. Two options exist: add a second reader for the HD21 layout, or extend the existing table format with a `jori` column and regenerate. The first is faster to get running; the second keeps one reader for both the shipped table and any future T-matrix computation, and is the better end state. Whichever is chosen, form

$$Q_{\rm pol}^{\rm abs}(\lambda, a) = \frac{1}{2}\left[Q_{\rm abs}^{(jori=3)} - Q_{\rm abs}^{(jori=2)}\right], \qquad Q_{\rm pol}^{\rm ext} \text{ likewise from } Q_{\rm ext}, \tag{24}$$

and watch the index convention of Table 1.

Carry it on the population. `grain_pop_t` (`sed/src/dust_model_mod.f90`) currently holds `aeff`, `dn`, `Cabs`, `Csca`, `kappB`, `H` and `kappCMB`. Add

```
real(wp), allocatable :: Cpol(:,:) ! (NLAM, NA) polarized absorption
real(wp), allocatable :: falign(:) ! (NA) alignment efficiency
```

and free them in `free_pop` alongside the rest. Filling `falign` from Eq. (9) needs only `aeff`, which the population already carries after the earlier fix that added it.

Accumulate the polarized emission. The emission loop in `sed_grain_loop` (`sed/src/sed_astrodust.f90`) has four accumulation sites, at roughly lines 757, 762, 926 and 938, of the form

```
Jout_local = Jout_local + dn_pop(ir) * P(ii) * Cabs_pop(:,ir) * spec
```

Each gains a companion line that differs only in the cross section and the alignment weight:

```
Jpol_local = Jpol_local + dn_pop(ir) * P(ii) * Cpol_pop(:,ir) &
                                * falign_pop(ir) * spec
```

This is the point made in §5.7: because the temperature weight multiplies both accumulators identically, the exact $P(T)$ treatment comes for free, and there is no reason to approximate the polarized emission with an equilibrium temperature. The two sites at 757 and 938 are the equilibrium branches and need the same companion.

Widen the public interface. `dust_emission` (`sed/src/sed_astrodust.f90:2143`) currently returns `lamI_total` and optionally `lamI_chan`. Add an optional `lamI_pol` of the same shape as `lamI_total`. Making it optional keeps every existing caller valid, which matters because MoCafe links this as a library; the same discipline was used for the `status` argument. Re-export through `dust_lib`.
Note that what leaves SEDust is the *intrinsic* polarized emissivity, that is, the quantity before the geometric factor $\sin^2\gamma$ and before any turbulent reduction. Those depend on the local field direction and belong to the RT side. Keeping the split here is what allows a single pair of tables to serve all viewing geometries.

Verify before going further. Reproduce `data/release/polarized_extinction.dat` by combining Eq. (5) with the alignment fit of Eq. (9) and integrating over the size distribution. This exercises the orientation indexing, the sign convention, the $f_{\rm align}$ fit and the size integration in one shot. It has already been shown to work at $0.55\,\mu$m; extending the check across the full wavelength grid is the natural first commit. Only then are the $p_{\rm submm}$, $R_{S/V}$ and Serkowski comparisons of §9. meaningful.

8.2 MoCafe, stage 2

A magnetic field has to exist. No field of any kind appears in the current source. A minimal implementation adds a field direction for each cell, either from an analytic model or from an MHD snapshot, alongside the existing density. For NGC 891 the analytic route is the starting point, but the observational result of Kim et al. (2023) means a turbulent component is not optional: a smooth field cannot produce the low polarization fraction and narrow dispersion they report. In practice this means either resolving the turbulence on the grid or carrying a reduction factor $F_{\rm turb}$ per cell, as GRADE-POL does.

Emissivity becomes a Stokes vector. `compute_dustemis` (`src/dustemis_mod.f90`) currently obtains a scalar spectrum from `dust_emission` or `dust_emission_interp` and stores one emissivity array per cell. With `lamI_pol` available it stores, for each cell,

$$j_I = (\text{as now}), \qquad \begin{pmatrix} j_Q \\ j_U \end{pmatrix} = (\lambda I_\lambda)_{\rm pol}\,\sin^2\gamma\,F_{\rm turb}\begin{pmatrix} \cos 2\varphi \\ \sin 2\varphi \end{pmatrix}, \tag{25}$$

where γ is the angle between the local field and the line of sight to the observer and φ is the position angle of the field projected on the sky. Both follow from the field direction and the observer direction, so no new physics enters here.
There is a structural consequence worth noting: γ and φ depend on the viewing direction, so j_Q and j_U are not properties of the cell alone in the way j_I is. For a single observer this is immaterial. For the all-sky output of `allsky_mod` it means the angles must be evaluated per direction rather than cached per cell.

Integrate the Stokes vector along the ray. The ray-tracing and peel-off paths (`src/peelingoff_mod.f90`, `src/allsky_mod.f90`) currently accumulate a scalar optical depth through `raytrace_tau` and apply $e^{-\tau}$. Replace this, for the polarized output only, by the segment-by-segment application of the closed-form solution of §5.: within a segment of constant density and field direction the Stokes vector transforms by cosh and sinh of $C_{\rm pol}ns$ in (I,Q) and by cos and sin of $C_{\rm circ}ns$ in (U,V), all scaled by $e^{-C_{\rm ext}ns}$, with a rotation into and out of the local field frame at each segment boundary. The existing segment walk already provides the geometry; what changes is what is done with each segment.

At this point the model produces polarized emission maps that can be compared with the HAWC+ 154 μm data directly. Nothing in the forward Monte Carlo has been touched.

8.3 MoCafe, stage 3

The forward Monte Carlo is where dichroism becomes awkward, because the optical depth that governs path sampling is no longer a scalar. The recommendation of §7. is to keep sampling path lengths from the scalar C_{ext} and to apply the polarization as a weight correction on the surviving packet, which preserves the sampling distribution and avoids the circularity that Baes et al. (2019) analyze. The physical cost is that dichroic extinction does not feed back on the radiation field; whether that matters can be measured by comparing stage-3 output with stage-2 output, which is a reason to build them in this order.

This is also the point at which `Galaxy_pol/dichroic.f90` would be retired. Its replacement is Eq. (2) with C_{pol} from Eq. (24), which restores the wavelength dependence that the empirical form cannot represent.

8.4 MoCafe, stage 4

Only here does the radiation-field tally change. `jtally_mod` stores `jt_sum(nlambda,ncell)`, a scalar mean intensity. Radiative torque alignment needs the direction of the radiation as well, so the tally becomes four numbers per cell and wavelength: the scalar sum and the three components of $\sum_i \vec{u}_{\lambda,i}$, from which the anisotropy degree γ_λ of Eq. (23) follows. The memory note already in the roadmap, that `jt_sum` is held privately per rank, becomes four times more pressing.

Two further constraints come with it. The modified random walk cannot be used in this pass, since it destroys the directional information. And the alignment solve itself is a preprocessing step over cells, not an inner-loop cost: as NewPOLARIS shows, the sensible structure is to compute a_{align}, a_{disr} and the magnetic radii once per cell, store them, and have the cross-section evaluation do nothing but compare grain size against stored radii.

8.5 Order of work and what each stage buys

Stage 1 is self-contained inside SEDust and verifiable against a published file. Stage 2 produces the first result that could not have been obtained in 2018, because the HAWC+ maps did not exist. Stage 3 removes the empirical formula that is the weakest point of the 2018 model. Stage 4 replaces the assumed $f_{\text{align}}(a)$ with computed alignment, and is the only stage that changes the Monte Carlo itself.

A useful property of this ordering is that each stage is independently publishable and independently falsifiable: stage 1 against `polarized_extinction.dat`, stage 2 against HAWC+, stage 3 against stage 2, and stage 4 against the fixed-f_{align} result of stage 2.

9. Validation targets

Correction to the classical limit. The last row deserves emphasis, because using the traditional value will let a bad model pass. The classical upper bound $p_V/E(B-V) \leq 9$ %/mag is obsolete; a model that under-polarizes will satisfy it

Table 3: Acceptance targets for a polarized dust model.

Quantity	Value	Source
$\lambda_{\max}$	$0.55\,\mu$m	`2009ApJ...696....1D`
Serkowski K	$-0.01+1.66\,\lambda_{\max}$	`1992ApJ...386..562W`
$p_{\max}$ at 353 GHz	$22.0^{+3.5}_{-1.4}$ %	`2020A&A...641A..12P`
$R_{S/V}$	4.2 ± 0.2	`2015A&A...576A.106P`
$R_{P/p}$	5.4 ± 0.2 MJy/sr	`2015A&A...576A.106P`
Polarization spectrum, $250\,\mu$m–3 mm	flat to $\lesssim 10\%$	HD23
$p_V/E(B-V)$	≥ 13 %/mag	`2020A&A...641A..12P`, `2019A&A...624L...8P`

comfortably. The post-Planck requirement is ≥ 13 %/mag, and the HD23 target value is 0.13/mag.

Values already computed here. Three numbers were obtained during this audit and are available as immediate checks:

- $Q_{\rm pol}/Q_{\rm abs} = 0.2790$ in the Rayleigh limit for $b/a = 1.4$, independent of size and wavelength. This is a pure geometry check and should be reproduced exactly.
- $p_{\rm submm} = 19.5\%$ with $f_{\rm align}$ applied, against the Planck value of 22.0%.
- $R_{S/V} = 3.85$ under a single-temperature approximation, against 4.2 observed and 4.31 as the HD23 target.

The third of these is the most informative, and it points to a natural first task. The 11% shortfall in $R_{S/V}$ is controlled by the relative emission weighting across grain sizes, and that weighting is exactly what SEDust's $T(a)$ and $P(T)$ solution provides. Recomputing $R_{S/V}$ with the full temperature treatment rather than a single T is therefore both a physics improvement and a sharp test of the new code path.

10. Unverified items

The following are stated as open, not as established. They are listed so that none of them is silently promoted to fact in later work.

1. $F_{\rm IR}$ **coefficient normalization.** Baseline POLARIS uses `1.40e10` where NewPOLARIS uses `6.14e9`. Whether this is a difference of unit system or a genuine disagreement was not determined. Since $F_{\rm IR}$ enters both Eq. (14) and Eq. (16), the factor matters and must be resolved before either is trusted.
2. **Origin of the $R_{S/V}$ gap.** Whether 3.85 versus 4.31 is entirely attributable to the single-temperature approximation, or partly to a difference in the definition of the ratio, is not established.
3. **Grain shape is an open choice, not a settled one.** Guillet et al. (2018) favor prolate grains with $a/b = 3$ plus porosity, while HD23 use a single-composition oblate spheroid with $b/a = 1.4$. These are mutually incompatible solutions to overlapping data, and adopting either is a modeling decision that should be recorded as such.

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